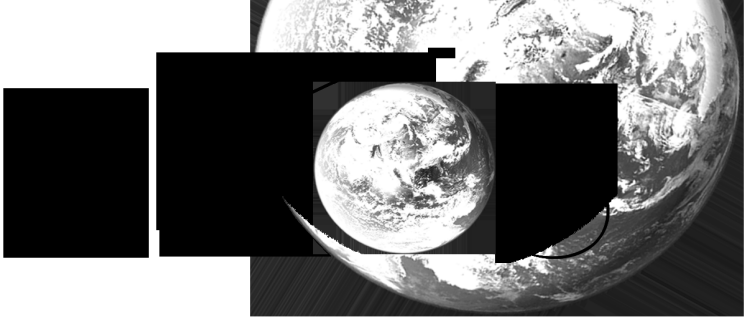


MEETING NOTES — NASA MSFC RDRE TECHNICAL DISCUSSION

October 6, 2026 | 1:00 PM CDT | Virtual

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1. Attendees

NASA MSFC

- Tom Teasley — RDRE Technical Lead, INROADS program, ER13 (combustion devices)
- Seth Bell — Industry & Commercial Partnerships Manager (also covers OGA / Other Government Agencies)
- Mike — Partnerships (surname unknown; Seth's colleague)
- Chris Hill — Project Manager / Approver for INROADS (not on call; must approve any SAA)

Exos Aerospace

- Jon Quinn — Cofounder & CEO
- Phil Gard — Chief Engineer (joined late, via phone)

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2. Background & Purpose

This was the first substantive technical-plus-business meeting between Exos Aerospace and NASA MSFC's RDRE team. Exos is conducting a trade-study down-select between (a) licensing the NASA MSFC INROADS RDRE design and (b) the GTL fully-machined RDRE path, for potential integration on a BLK3-class vehicle flight demonstration. This meeting opened both the technical and agreement-structure discussions.

> *Transcription note: 'INROADS' (or 'in roads') appears to be the NASA MSFC RDRE development program name. '9,000–12,000 kilohertz' in transcript is almost certainly 9–12 kHz.*

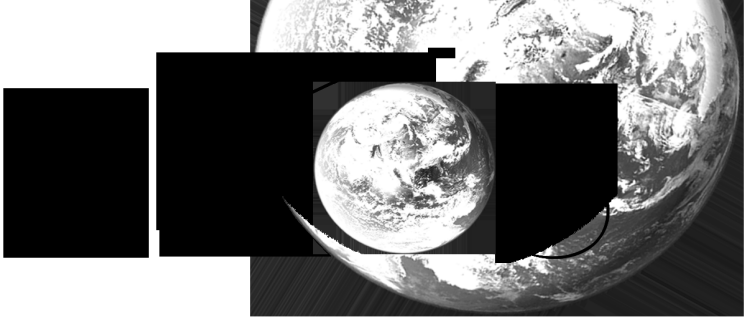
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3. Technical Facts (per Teasley, verbal — 2026-10-06)

3.1 Engine Performance & Tested Hardware

- Maximum tested thrust: 5,800 lbf (sea level, pressure-fed to date). (per Teasley, verbal)
- Baseline design: ~20 lb/s mass flow → ~5,000–6,000 lbf at sea level. (per Teasley, verbal)
  - At altitude (~10,000 ft) same conditions yield ~8,000 lbf+. (per Teasley, verbal)
- Baseline operating conditions: ~1,000 psi chamber pressure / ~1,700 psi inlet pressure. (per Teasley, verbal)





• Throttle: baselined 50%; possibly 40% (thrust chamber only; other components may limit further). (per Teasley, verbal)

• Cooling: all regenerative — no film cooling used. (per Teasley, verbal)

• Notional 10,000-lbf engine system: baselined for turbo-pump-fed methane-rich gas generator cycle (50–100% = ~5,000–10,000 lbf). (per Teasley, verbal)

3.2 Efficiency

• ≥98% combustion efficiency at ≥700 psi chamber pressure. (per Teasley, verbal)

• ≥90% combustion efficiency above ~300 psi chamber pressure. (per Teasley, verbal)

• ISP ~250 s at ~300 psi Pc / 20 lb/s / ~5,000 lbf (LOX-methane or LOX-ethanol near-equivalent). (per Teasley, verbal)

3.3 Loads Environment

• Measured loads: up to ~1,000+ g in some test configurations. (per Teasley, verbal)

• Dominant frequency range: ~9–12 kHz. (per Teasley, verbal)

Transcript reads '9,000–12,000 kilohertz' — almost certainly 9–12 kHz.

• Loads data book: not yet released — data scattershot, not yet interpretable. (per Teasley, verbal)

• Early: metal C-rings shattered; pushing frequency higher → smaller items (instrumentation ports) failing. (per Teasley, verbal)

• No chamber wall erosion observed in any test. (per Teasley, verbal)

• Detonations localized within ~0.25 inch of injector face. (per Teasley, verbal)

• Resonator cavity testing successfully mitigated targeted frequencies. (per Teasley, verbal)

3.4 Inlet Pressure / Exos Interface Compatibility

• 750 psi inlet: requires outer-body redesign + remanufacture to open throat, reducing Pc/inlet P. (per Teasley, verbal)

• ~25 lb/s for ~6,000 lbf SL → ~1,200 psi inlet required — pushes toward pump augmentation. (per Teasley, verbal)

• LOX/CH4 turbo-pump <10,000 lbf: none found in NASA repository. (per Teasley, verbal)

• Ursa Major Hadley: ~5,000 lbf LOX/kerosene oxy-rich — turbo-pump possibly adaptable with RDRE chamber. (per Teasley, verbal)

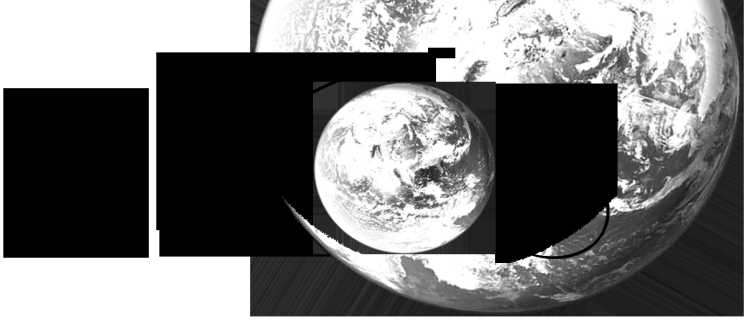
3.5 Hardware Availability & Timeline

• Existing non-flight-weight hardware: available now; no performance or duration guarantee. (per Teasley, verbal)

• Higher-confidence hardware: post-1B (≈Mar–May 2027) / post-1C series — likely end-2027 at earliest. (per Teasley, verbal)

• New thrust chamber (license current design): ~12–18 months. (per Teasley, verbal)





- Exos-owned design with Teasley coaching: approximately 2 years. (per Teasley, verbal)
- Turbomachinery: several years. (per Teasley, verbal)
- Additive timeline: ~10 days print + ~3 wks depowder + optional CT scan + HIP ≤2 wks + flow test ≤2 wks. (per Teasley, verbal)
- GRCop powder supply: easing — primary suppliers reportedly restoring quotas. (per Teasley, verbal)
- Fastest path: license current INROADS design. (per Teasley, verbal)

3.6 Vacuum / In-Space Operation

- Restart in vacuum: possible — MSFC work done, details not releasable. (per Teasley, verbal)
- No funded flight demo in INROADS program; any flight demo 5–10+ years under NASA's plan. (per Teasley, verbal)

Jon stated Exos could potentially do this in ≤2 years — noted as Exos aspiration, not a NASA commitment.

3.7 INROADS Program IP / Publications

- Heat-transfer correlation paper: in journal review (JNF/JANNAF) — one equation predicts convective heat transfer coefficient for LOX/CH4 RDE hot wall. (per Teasley, verbal)
- JPP paper: design and prediction of wave count and speed — validated vs. ~300 hot fires. (per Teasley, verbal)

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4. Proposed Agreement Path (per Teasley / Mike / Seth, verbal)

- Step 1: Exos provides firm written requirements document.
- Step 2: Teasley socializes with PM Chris Hill → Hill must approve.
- Step 3: Umbrella Reimbursable Space Act Agreement (SAA) with annexes.
  - Annex 1: paid consultation (Exos pays for Teasley's time; Seth/Mike overhead / no direct cost).
  - Subsequent annexes added as scope evolves.
- Licensing path: Exos has NOT engaged NASA Tech Transfer — must do so separately for an IP license.

All SAAs are public documents — searchable on NASA.gov. Exos has 1–2 existing SAAs.

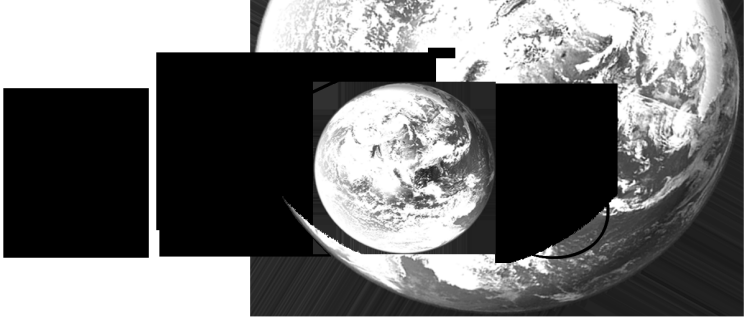
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5. Exos-Stated Intents (Jon / Phil, on call — pre-decisional)

The following reflects what Jon and Phil stated on the call. Pre-decisional; subject to board approval and trade-study completion.

- Objective: hover then fly — generate in-flight data on isolation/damping requirements.
- Burn duration: ~85 seconds.
- Propellant: LOX/CH4 or LOX/ethanol (tank sizing ~1:1).





- Thrust target: ~6,000 lbf sea level.
- Tank inlet options Exos can supply: 750 / 1,000 / 1,200 psi (CF tanks manufactured in-house).
- Approximate program budget mentioned: ~$10M for a BLK3-class flight demo.
- Exos previously flew a NASA active vibration damper — familiar with isolator/damper hardware.

■ **INTERNAL ONLY — Do NOT describe Exos as an 'RDRE propulsion provider' or as offering propulsion packages. RDRE is a trade-study down-select (MSFC license vs. GTL machined). Exos does not own MSFC RDRE technology.**

6. Open Questions

- Exact Exos inlet pressure requirement — drives outer-body redesign decision.
- Propellant selection: LOX/CH4 vs. LOX/ethanol.
- Phil to confirm mass flow and thrust precision vs. Teasley's 20 lb/s baseline.
- Can ~6,000 lbf be achieved at 750 psi inlet without outer-body redesign?
- Electric pump vs. clean-sheet design path at 5,000–6,000 lbf — cost/schedule?
- NASA Tech Transfer contact / license path — who initiates and timeline?
- 'RCG' data meaning — confirm with Dankanich.

7. Actions

A-1 [Jon]: Draft and finalize firm written RDRE requirements; send to Teasley for Chris Hill review. [Decision posted 2026-10-07]

A-2 [Phil]: Confirm precise mass flow, inlet pressure, and thrust requirement vs. Teasley's 20 lb/s baseline.

A-3 [Jon]: Initiate NASA Tech Transfer contact for INROADS license path.

A-4 [Jon]: Update Dankanich on 10/6 meeting outcome; advance GRCop/PSI thread.

A-5 [Seth Bell — MSFC]: Prepare umbrella SAA framework for when requirements are ready.

Exos Aerospace Systems & Technologies, Inc. | 299 FM 1903, Greenville, TX 75402 | jquinn@exosaero.com

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